Algebra 1 Name____________________
Unit 2 Date____________________
Lesson 1 Practice Period____________________

1. The power of a power law of exponents states to keep the
   - ○ base
   - ○ coefficient
   - ○ exponent

   and then
   - ○ add
   - ○ subtract
   - ○ multiply

   the powers.

2. Diana simplified the algebraic expression and made an error. Find and fix the mistake.

$$(-3x^3y^5)^3(1.5xy^0)$$
$$(-3x^9y^{15})(1.5x)$$
$$(-3y \cdot 1.5)(y^{15})(x^9 \cdot x)$$
$$-4.5x^9y^{15}$$

3. Write a note to Diana explaining the error she made in the previous question.

Determine an equivalent algebraic monomial expression for each expression using the laws of exponents. Show your work.

4. $(-8a^5b)(3ab^4)$

5. $(\frac{-1}{4}x^3y^2)(4x^8y^3)(-5y^2)$

6. $(2.8kj^9)(3.5k^8)\left(\frac{1}{3}k^5j^2\right)$

7. $(-3mn^5p^4)^3$

8. $(2.3c^4d)(-2.4c^{-3}d^0)(-c^2d^{-5})$

9. $\left(\frac{5}{6}r^{-8}\right)^{-2}(-9.6r^5s^4)^0$

10. $(-8y^5z)^3(5y^7z^6)^3$